A low-current verification benchmark for two-dimensional MOSFET drift–diffusion simulation: reference conductances, failure signatures of terminal-current evaluation, and an AI-agent study

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Abstract

Terminal currents are what device simulators are judged by, yet at low current they are often computed from a flux sum at a heavily doped contact, where drift and diffusion nearly cancel. We release an open, code-agnostic benchmark for the zero-bias (linear-response) terminal conductances of two-dimensional n-channel MOSFETs with resolved n^+ wells in deep subthreshold, $\hat{\psi} \approx 12$–21, where the drain current at 1 mV ranges from 1.7×10^{-15} to 1.6×10^{-8} A/cm. The reference values are the terminal currents of the linear-response Scharfetter–Gummel network evaluated in cancellation-free (weighting-function) form. They are cross-checked by an independent finite-volume solver and by drift–diffusion runs, under acceptance criteria fixed before the runs: all 6 primary states and all 5 held-out states pass. The primary states can be scored with any code on any mesh, to a stated tolerance of 2.4×10^{-4}–4.7×10^{-3}, or on a published mesh to 10^{-6}. With the benchmark we document two failure signatures in DEVSIM 2.11.2. Its default post-solve contact current has an absolute round-off floor of 2.4×10^{-12}–2.7×10^{-11} A/cm and fails all 17 scorable cases; extended precision repairs it at 10.9–15.7 times the cost. Its built-in AC analysis returns the coupled small-signal solve, whose backward error is near machine precision while its output is wrong by up to 3.3×10^3 relative, in double and in extended precision. Solving the same double-precision system exactly does not repair it: rounding the Jacobian entries by one unit in the last place moves the output by up to 93% of the error. An exploratory analysis traces this to the neutral drain, where the potential derivatives of the fluxes cancel in exact arithmetic and, once rounded, leave an error floor of 3.2×10^{-9}–8.6×10^{-9} S/cm that does not depend on the gate bias. The references converge at second order under mesh refinement. Finally, in a pre-registered study an AI agent (Claude Opus 5.5 in Claude Code) computed the conductance in 33 independent sessions without the scorer. It produced no silent error (0/33; 95% upper bound 0.106): every session called DEVSIM's contact current, most also its AC analysis, and all set them aside after independent checks. The registered stop rule therefore ended the study before its second arm. All data, code and transcripts are released.

1 Introduction

A device simulator is judged by its terminal currents. Below threshold and at a drain bias of one millivolt, the drain current of the devices studied here is as small as 10^{-15} A per centimetre of width, while the electron density at the contacts is 10^{20} cm^{-3}. If the current is computed as the flux through the contact boundary, the result is the small difference of two large terms, drift and diffusion, and its accuracy is limited by round-off rather than by the discretization. The loss of accuracy of flux integration at low current in heavily doped regions has been known since

at least the early 1980s; Palm and Van de Wiele analysed it, attributed the first observation to earlier work cited there, and proposed a remedy based on a current potential [1]. A related family of remedies evaluates the terminal current with a weighting function, an idea that goes back to the Ramo–Shockley theorem [2, 3] and includes the characteristic-potential method of noise analysis [4] and the test-function current used by Farrell and Peschka [5].

Verification of a simulation result needs reference solutions with stated uncertainty [6, 7]. For drift–diffusion discretizations an open benchmark exists: Farrell and Peschka compare finite-difference, finite-element and Scharfetter–Gummel finite-volume schemes on forward-biased p-i-n diodes in one and two dimensions and release their data [5]. Cross-method comparisons have also been published for kinetic transport [8] and for formulations and nonlinear solvers [9], and discretizations other than the vertex-based Scharfetter–Gummel scheme are in use [10]. In the literature we reviewed we did not find an open reference set for the low-current, linear-response terminal conductance of a two-dimensional MOSFET, the regime in which flux integration fails. We state this narrowly: it is the result of a search, not a proof of absence.

A second motivation is new. Large language model (LLM) agents now write and run TCAD decks [11, 12], and agents in general are known to report success on wrong results ("silent" or "false success" failures) [13, 14]. Pairing an agent with a verifier reduces such failures in scientific simulation [15, 16]. A low-current terminal current is a natural test case: the dangerous errors are physically known, and they leave the solver residual clean, so a check of convergence alone does not reveal them.

This paper makes three contributions; the new runs behind each were pre-registered before they were made.

1. **A benchmark** (Sections 2 to 4): 106 case files, of which 6 primary three-terminal states (tier A-R) carry a tolerance usable on any mesh and with any code, together with published meshes, a JSON schema, a scorer and worked examples.

2. **A failure-mode atlas** (Section 5): quantified failure signatures of common evaluations of the terminal current, including two user-facing paths of the open simulator DEVSIM [17, 18], and their remedies. The flux-integration failure is classical [1]; the failure of the coupled small-signal solve, with a backward error near machine precision, was not found in the literature we reviewed. We trace it to the rounding of the Jacobian entries in the neutral drain, which no double-precision solver can undo.

3. **An AI-agent study** (Section 6): a pre-registered test of whether an LLM agent reports a wrong conductance as correct when it has no access to the scorer. The answer, for the agent and conditions tested, is no.

We make no claim of novelty for the reference method itself: it is the established weighting-function evaluation. Everything reported is agreement between independent computational paths; nothing here is experimental validation.

2 Problem definition

2.1 Model and scope

All cases share one physical model (Table 1). In silicon,

$$\nabla \cdot (\varepsilon_{\mathrm{Si}} \nabla \psi) = -q\,(p - n + N_D - N_A), \qquad \nabla \cdot \mathbf{J}_n = 0, \qquad \mathbf{J}_n = -q\mu_n n \nabla \phi_n, \tag{1}$$

with $n = n_i \exp[(\psi - \phi_n)/V_T]$ and holes held at equilibrium, $p = n_i \exp(-\psi/V_T)$; in the oxide $\nabla \cdot (\varepsilon_{\mathrm{ox}} \nabla \psi) = 0$. The potential ψ is referenced to the intrinsic level. Source and drain are ohmic ($\psi = V_T \operatorname{asinh}(N_{\mathrm{net}}/2n_i) + V$, $\phi_n = V$), the gate sets $\psi = V_G + \psi_{b0}$ on the oxide top (flat-band voltage zero), and the remaining boundaries carry zero normal displacement and zero normal

Table 1: Scope envelope shared by every case.

Statistics	Boltzmann
Mobility	constant, $\mu_n = 1350\ \mathrm{cm}^2/(\mathrm{V\,s})$
Holes	at equilibrium, not part of the observable
Generation/recombination	none
Temperature, intrinsic density	$T = 300$ K, $n_i = 10^{10}\ \mathrm{cm}^{-3}$
Permittivities	$\varepsilon_{\mathrm{Si}} = 11.7\,\varepsilon_0$, $\varepsilon_{\mathrm{ox}} = 3.9\,\varepsilon_0$
Gate	$\psi_{\mathrm{gate}} = V_G + \psi_{b0}$ (no work-function difference, no oxide charge)
Bias	equilibrium; observables are $V_D \to 0$ limits

electron current. The body contact at the bottom of the silicon has two variants: ohmic (B1: $\psi = \psi_{b0}$, $\phi_n = 0$) and electron-blocking (B0: $\psi = \psi_{b0}$, zero normal electron current). Currents are per unit width (A/cm) and conductances in S/cm.

2.2 Linear response at zero drain bias

At $V_D = 0$ the device is in equilibrium: $\phi_n \equiv 0$ and $n = n_0(x, y)$. Write $\phi_n = V_D\,\varphi + O(V_D^2)$. Because $\nabla\phi_n = O(V_D)$, the change of n enters $\mathbf{J}_n$ only at second order, so to first order

$$\mathbf{J}_n = -V_D\,\sigma_0 \nabla\varphi, \qquad \nabla\cdot(\sigma_0\nabla\varphi) = 0, \qquad \sigma_0 = q\mu_n n_0, \tag{2}$$

with $\varphi = 1$ on the drain, $\varphi = 0$ on the source (and on the body for B1), and zero normal flux elsewhere. The electrostatic response does not enter at this order. The drain and source conductances follow as

$$G_{\mathrm{DD}} \equiv \frac{\partial I_D}{\partial V_D} = \int_\Omega \sigma_0\,|\nabla\varphi|^2\,\mathrm{d}A, \qquad G_{\mathrm{SD}} \equiv -\frac{\partial I_S}{\partial V_D}, \qquad G_{\mathrm{DB}} \equiv -\frac{\partial I_B}{\partial V_D}, \tag{3}$$

with $G_{\mathrm{DD}} = G_{\mathrm{SD}} + G_{\mathrm{DB}}$. The function φ is the weighting function of the drain: in the language of noise analysis it is the characteristic potential at zero frequency [4], and the energy form in Eq. (3) is the linear-response case of the test-function current [5], an evaluation that does not subtract at the excited contact. Two-terminal cases (body B0) have $G_{\mathrm{DD}} = G_{\mathrm{SD}} \equiv G_0$. Every simulator can produce these quantities as the $V_D \to 0$ limit of its terminal currents, which is how the benchmark scores them.

We also report the per-end correction $\Delta L = (L/2)(1 - G_{\mathrm{GCA}}/G_{\mathrm{SD}})$, where G_{GCA} is the long-channel (one-column) sheet conductance divided by L; it expresses the two-dimensional end effect as a length.

2.3 Discrete reference

On a mesh, box integration with Scharfetter–Gummel fluxes [19] turns Eq. (2) into a resistor network with edge conductances

$$c_e = q\mu_n\,\frac{A_e}{h_e}\,n_{0,a}\,B\!\left(\frac{\psi_a - \psi_b}{V_T}\right), \qquad B(x) = \frac{x}{e^x - 1}, \tag{4}$$

where h_e is the length of edge $e = (a, b)$ and A_e its box-face coupling; c_e is symmetric in a and b at equilibrium. With φ the network solution, the reference values are $G_{\mathrm{DD}} = \varphi^{\mathsf{T}}\mathcal{L}\varphi = \sum_e c_e(\varphi_a - \varphi_b)^2$ (edge energy; second form $-(I_S + I_B)$), $G_{\mathrm{SD}} = -I_S$ (second form: reciprocity, $-I_D$ with the source excited), and $G_{\mathrm{DB}} = -I_B$. Every reference form is evaluated where $\varphi = 0$ or as an energy, never as a sum at the excited contact. For two-terminal cases the network on a given mesh is bracketed by Rayleigh's monotonicity law [20]: shorting every mesh column gives an upper bound G_{slice}, cutting every vertical edge a lower bound G_{rows}. The bracket bounds that discrete network, not the continuum value.

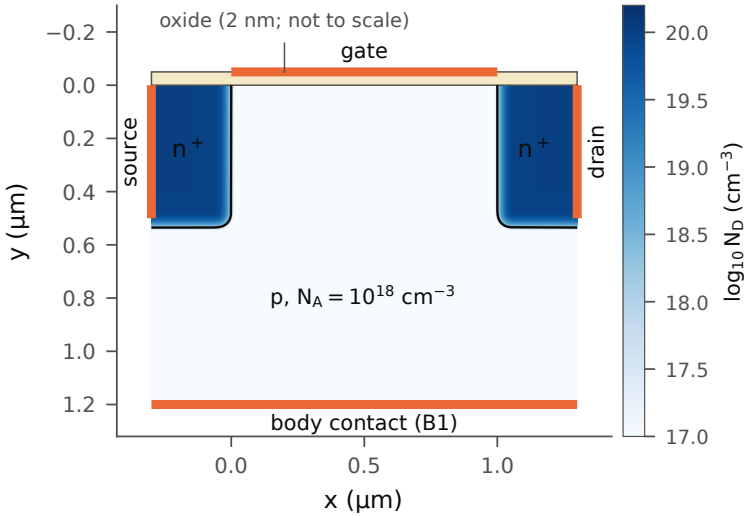


Figure 1: Deck R, device dev1: donor profile of the resolved n^+ wells (shading; black line: metallurgical junction $N_D = N_A$) and the contacts (orange). The oxide is drawn thicker than its 2 nm. Body contact B1 is ohmic; its two-terminal companion B0 blocks electrons.

Table 2: Benchmark tiers. $\tau_{\mathrm{num}} = 10^{-6}$ relative is the same-mesh tolerance.

Tier	Cases	Content	Scored by
A-R	6	deck R, body B1, G_{DD}, G_{SD} (G_{DB} reported)	tolerance on any mesh; τ_{num} on a published mesh
HO	5	held-out deck-R states, level-1 mesh	τ_{num} on the published mesh
B-R	23	body-B0 companions of every A-R and HO state	same-mesh Rayleigh bracket
A-W	3	deck W (internal)	tolerance and same-mesh bracket
B-W	28	deck-W deep-subthreshold states	same-mesh bracket
C	41	stress: $V_G \geq 1$ V, well depth, finite bias	where established

2.4 Devices, states and tiers

The primary device (deck R, Fig. 1) resolves the n^+ source and drain wells: silicon spans $-0.3 \leq x \leq L + 0.3$ μm and $0 \leq y \leq 1.2$ μm; the wells are analytic erfc profiles with peak 10^{20} cm^{-3}, metallurgical junctions at $x = 0$ and L ($\sigma = 15$ nm) and depth $h_w = 0.5$ μm. Two devices are used: dev1 ($N_A = 10^{18}$ cm^{-3}, $t_{\mathrm{ox}} = 2$ nm, $L = 1$ μm) and dev2 ($N_A = 10^{16}$ cm^{-3}, $t_{\mathrm{ox}} = 20$ nm, $L = 2$ μm). Gate voltages are chosen by the long-channel band bending $\hat{\psi} = (\psi_{s,\infty} - \psi_{b0})/V_T$ at $\hat{\psi} \approx 12$, 16 and 21 for the primary tier, and 15, 18 (and 12 at $L = 2$ μm) for the held-out tier, all below threshold, from depletion to weak inversion. Each case file contains the complete device specification, so a case can be set up without the code that produced it.

Table 2 lists the tiers. Tier A-R and the held-out tier HO are the external-facing three-terminal sets; B-R are their electron-blocking companions, one case per mesh. Tiers A-W and B-W use an internal, truncated deck W in which the wells are replaced by a Dirichlet condition on a vertical segment at the well edge; standard TCAD codes cannot reproduce it, and these tiers are kept for codes that can. Tier C holds stress cases (gate voltages of 1 V and above, finite drain bias, well depth) with cross-path agreement only.

2.5 Meshes

Every reference mesh is a published tensor-product mesh whose rectangles are split into two triangles by a stated rule; the node lines are given exactly (`meshes/*.json`), so a box-integration Scharfetter–Gummel code can rebuild the identical network and declare a same-mesh submission.

Three levels are published per device; the level-1 meshes have 368×119 (dev1) and 438×315 (dev2) silicon nodes, the level-2 meshes 521×167 and 621×445.

3 Reference values and their uncertainty

3.1 Three computational paths

P1, network. The equilibrium of Eq. (1) is solved with DEVSIM [17] by Newton's method on the published mesh. The network of Eq. (4) is assembled from that equilibrium, solved by sparse LU factorization, and evaluated in the forms of Section 2.3, at levels 0, 1 and 2. **P4, independent finite volume.** A separate code that imports only NumPy and SciPy, with its own tensor mesh, box-integration assembly, damped-Newton equilibrium and network. It shares no code with P1, but it was written by the same author for the same problem statement: it is independent in implementation, not in authorship. **P2, drift–diffusion.** DEVSIM solves the full coupled problem at $V_D = \pm 0.1$ and ± 1 mV with a circuit voltage source on the drain. The current is taken in three ways: DEVSIM's post-solve contact current, the current of the circuit voltage source, and a weighting-function current. Central differences $G(\delta) = [I(\delta) - I(-\delta)]/2\delta$ are combined as $G_R = [100\, G(0.1\ \text{mV}) - G(1\ \text{mV})]/99$.

3.2 Pre-registered acceptance

The primary states, the held-out states and every threshold were fixed before the first production run; amendments were dated and appended, and none changed a state, threshold or tolerance rule. A primary state (H1) had to satisfy, for G_{DD} and G_{SD}, $|P4/P1_{L2} - 1| \le 5 \times 10^{-3}$, $|P1_{L2}/P1_{L1} - 1| \le 5 \times 10^{-3}$, agreement of the two forms and current conservation within 10^{-9}, and the B0 bracket at every level; the tier was to be withdrawn if fewer than 4 of 6 states passed. All 6 passed; the largest P4 difference was 2.3×10^{-3} and the largest level-2 to level-1 change 2.3×10^{-3}. All 5 held-out states passed their criteria (forms and conservation within 10^{-9}, bracket). In the drift–diffusion runs, the circuit-node current's Richardson value agreed with the P1 network within 1.2×10^{-9} in all 23 runs.

3.3 Tolerances

The reference value of an A-R state is P1 on the level-2 mesh. Its tolerance, applied separately to G_{DD} and G_{SD}, is $2\,(u_{\mathrm{disc}} + u_{\mathrm{round}} + u_{\mathrm{solver}})$, where $u_{\mathrm{disc}} = \max(|P1_{L2}/P1_{L1} - 1|, |P4/P1_{L2} - 1|)$, u_{round} is the spread of the two forms and u_{solver} the network residual. It ranges from 2.4×10^{-4} to 4.7×10^{-3} (Table 3). On a published mesh the same-mesh reference is exact to round-off and a submission is held to $\tau_{\mathrm{num}} = 10^{-6}$. Because the P4 spread is part of the tolerance, P4 passes by construction: that is a consistency check, not independent validation. Figure 2 shows the range of the references.

Grid convergence. A three-grid analysis of the P1 values at levels 0, 1 and 2 [21, 22], with the refinement ratio taken from the node counts (about 1.41 per level), supports these tolerances (Table 4). It was pre-registered with three predictions, all of which hold. Convergence is monotone in all 12 of 12 cases (six states, G_{DD} and G_{SD}), with observed orders 1.98–2.22, as expected of a second-order scheme. Farrell and Peschka found second-order convergence of the Scharfetter–Gummel finite-volume current for smooth doping, and only first-order convergence of their finite-volume potential for near-discontinuous doping [5]; our wells have smooth erfc profiles. The asymptotic-range ratio is 0.998–1.000 (1 in the asymptotic range). The level-2 references lie 1.2×10^{-4} to 2.0×10^{-3} from the Richardson-extrapolated values; the registered tolerances contain these differences by a factor of 2.0 to 2.3, and contain the grid convergence index of the finest mesh (1.4×10^{-4} to 2.5×10^{-3}) in 12 of 12 cases. In 6 of 6 states the independent

Table 3: Three-terminal reference values (body B1). A-R values are P1 at level 2, with the relative tolerance for any mesh; HO values are the level-1 same-mesh references.

State	Device	V_G (V)	$\hat{\psi}$	G_{DD} (S/cm)	G_{SD} (S/cm)	ΔL (nm)	Tolerance
AR-1	dev1	0.488174	12.0	3.5276×10^{-12}	3.5041×10^{-12}	34.21	2.8×10^{-3}
AR-2	dev1	0.621433	16.0	1.6162×10^{-10}	1.6160×10^{-10}	38.52	3.7×10^{-3}
AR-3	dev1	0.7835	21.0	2.1098×10^{-8}	2.1098×10^{-8}	40.84	4.7×10^{-3}
AR-4	dev2	0.488174	12.0	2.5502×10^{-9}	2.5451×10^{-9}	364.23	2.6×10^{-4}
AR-5	dev2	0.621433	16.0	1.2034×10^{-7}	1.2033×10^{-7}	381.53	2.5×10^{-4}
AR-6	dev2	0.7835	21.0	1.5974×10^{-5}	1.5974×10^{-5}	395.18	2.4×10^{-4}
HO-1	dev1	0.588534	15.0	6.1776×10^{-11}	6.1753×10^{-11}	37.92	τ_{num} only
HO-2	dev1	0.686557	18.0	1.1211×10^{-9}	1.1211×10^{-9}	39.54	τ_{num} only
HO-3	dev2	0.588534	15.0	4.5692×10^{-8}	4.5687×10^{-8}	377.87	τ_{num} only
HO-4	dev2	0.686557	18.0	8.4012×10^{-7}	8.4011×10^{-7}	388.06	τ_{num} only
HO-5	dev1, $L = 2\,\mu$m	0.488174	12.0	1.7171×10^{-12}	1.6840×10^{-12}	29.38	τ_{num} only

Table 4: Grid convergence of the A-R references (G_{DD}; G_{SD} agrees to two digits): observed order p, distance of the level-2 value from the extrapolated value, grid convergence index of the finest mesh (safety factor 1.25), registered tolerance, and distance of the independent finite-volume value (P4) from the extrapolated value.

State	p	$\lvert G_{L2}/G_{\mathrm{ext}} - 1\rvert$	$\mathrm{GCI}_{\mathrm{fine}}$	Tolerance	$\lvert G_{P4}/G_{\mathrm{ext}} - 1\rvert$
AR-1	2.21	1.3×10^{-3}	1.6×10^{-3}	2.8×10^{-3}	1.6×10^{-4}
AR-2	2.22	1.6×10^{-3}	2.0×10^{-3}	3.7×10^{-3}	2.3×10^{-4}
AR-3	2.21	2.0×10^{-3}	2.5×10^{-3}	4.7×10^{-3}	2.9×10^{-4}
AR-4	2.02	1.2×10^{-4}	1.4×10^{-4}	2.6×10^{-4}	1.5×10^{-5}
AR-5	2.00	1.2×10^{-4}	1.5×10^{-4}	2.5×10^{-4}	6.0×10^{-6}
AR-6	1.98	1.2×10^{-4}	1.5×10^{-4}	2.4×10^{-4}	1.9×10^{-5}

finite-volume value lies closer to the extrapolated value (within 2.9×10^{-4}) than the finest network value does.

4 Scorer and submission format

A submission is a JSON file that maps case identifiers to values (`G_DD_S_per_cm`, `G_SD_S_per_cm`, or G_0) and may declare `same_mesh` with a published mesh identifier. The scorer applies, in order: a sign check (a conductance ≤ 0 fails); for a same-mesh three-terminal submission, $|G/G_{\mathrm{ref}} - 1| \leq \tau_{\mathrm{num}}$ against the reference on that mesh; for any other A-R submission, the tolerance of Table 3 and the matching ΔL tolerance; for two-terminal same-mesh submissions, the Rayleigh bracket with a relative slack of 10^{-12}. A case with no applicable check is reported as not scorable. Table 5 lists the worked examples shipped with the benchmark: each re-submits an evaluation path from the production records through the unchanged scorer.

5 Failure-mode atlas

Figure 3 summarizes the error of each evaluation path against the same-mesh reference on the level-1 mesh, as a function of the size of the current. The scope of every statement in this section is these devices, these meshes and DEVSIM 2.11.2.

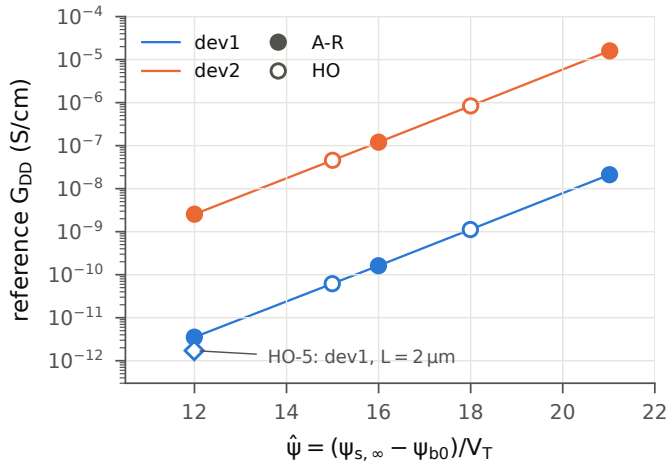


Figure 2: Reference G_{DD} against the long-channel band bending $\hat{\psi}$ for dev1 and dev2 (filled: A-R; open: HO). The drain current at 1 mV is $G_{\mathrm{DD}} \times 1$ mV, from 1.7×10^{-15} to 1.6×10^{-8} A/cm.

Table 5: Worked examples: evaluation paths submitted to the scorer (cases passed and failed).

Submitted evaluation path	Pass	Fail
Reference forms (A-R, HO), same mesh	11	0
DEVSIM circuit-node current $-x$[V1.I], Richardson, same mesh	11	0
Independent finite volume (A-R), other mesh	6	0
DEVSIM `get_contact_current`, double, Richardson, same mesh	0	11
DEVSIM `get_contact_current`, extended precision (A-R)	6	0
Coupled small-signal solve G_E (full Jacobian), level-1 mesh	0	11
DEVSIM AC analysis, double (A-R)	0	6
DEVSIM AC analysis, extended precision (A-R)	0	6
Two-terminal source-side reference, same mesh	86	0
Two-terminal drain-side flux sum, same mesh	48	38
Two-terminal coupled small-signal G_E, same mesh	0	14
Two-terminal independent finite volume (A-W), other mesh	3	0

5.1 Flux sum at the excited contact

Summing the edge currents at the drain contact of the network itself, rather than using an energy or source-side form, misses the same-mesh reference by more than τ_{num} on every A-R and HO mesh, and by up to 4.0×10^3 relative, with the wrong sign, at $\hat{\psi} = 12$ (Fig. 3a, crosses). The source-side sum of the same solution is exact to round-off. The error is a round-off effect, not a discretization error: it does not decrease under mesh refinement, while the source-side and energy forms of the same solution agree to 10^{-9} or better. We did not analyse its mechanism further.

5.2 DEVSIM's post-solve contact current

A related failure appears in DEVSIM's default contact current (`get_contact_current`) of a full drift–diffusion solution. Its absolute error, measured against the circuit-node current of the same runs, was 2.4×10^{-12} to 2.7×10^{-11} A/cm on these meshes. On the dev1 level-1 mesh at $V_D = +0.1$ mV the raw drain current was 1.44×10^{-11} A/cm, identical to all digits, in 5 runs (AR-1, AR-2, AR-3, HO-1, HO-2) whose G_{DD} spans 3.5×10^{-12} to 2.1×10^{-8} S/cm: below about 10^{-11} A/cm the value carries no channel information. Pre-registered predictions of the error at ± 1 mV (above 10^{-2} where $G_{\mathrm{DD}} \times 1$ mV $\leq 10^{-10}$ A/cm, below it where $\geq 10^{-8}$ A/cm) were confirmed for all 9 predicted states, and, scored as a submission, this current failed all 17 scorable cases (Fig. 3a, triangles). Two remedies pass: the circuit-node current, whose

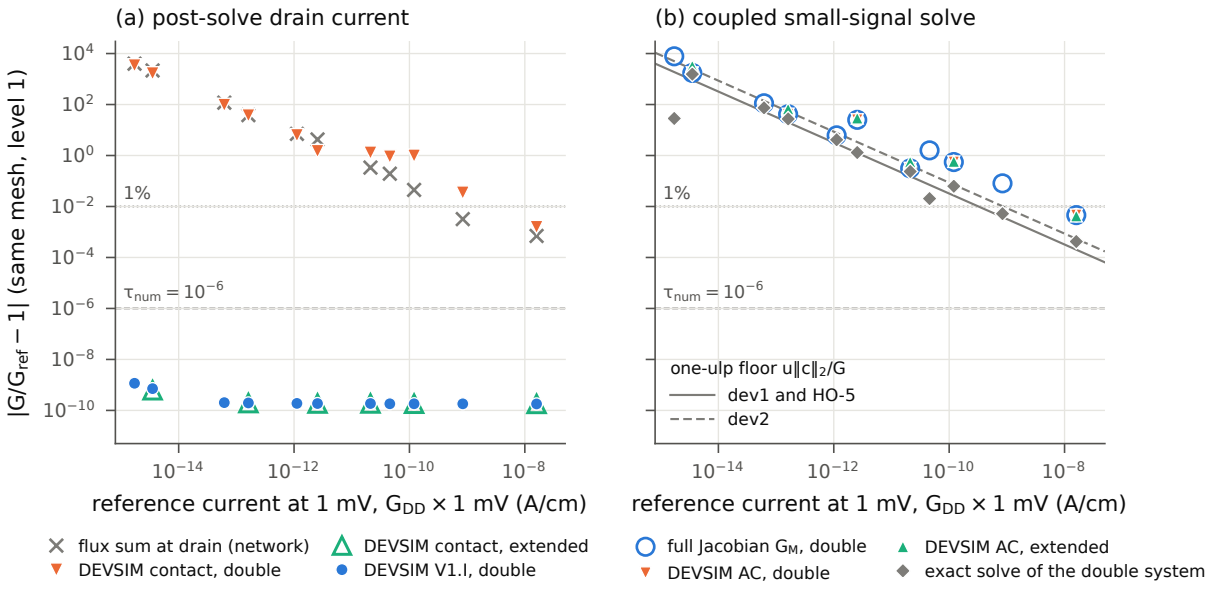


Figure 3: Relative error of G_{DD} from each evaluation path, against the size of the current, for the 11 A-R and HO states on their level-1 meshes (body B1). (a) Post-solve currents, Richardson values G_R of the drain current (DEVSIM paths), and the drain-side flux sum of the network itself. (b) The coupled small-signal solve of the full Jacobian, DEVSIM's AC analysis (A-R states), and the exact solution of the same double-precision system (diamonds, all 11 states). Gray lines: the one-ulp data floor $u\|c\|_2/G$ of each device (exploratory analysis, Section 5.3); HO-5 shares the floor of dev1. Dashed: τ_{num}; dotted: 1%.

Richardson value matched the reference within 1.2×10^{-9} in all runs, and DEVSIM's extended precision (`extended_solver`, `extended_model`, `extended_equation`), with which the contact current equalled the circuit-node current to double-precision resolution and its Richardson value matched the reference within 6.2×10^{-10} in 6 of 6 A-R states, at 10.9–15.7 times the wall time.

5.3 The coupled small-signal solve and DEVSIM's AC analysis

The linear-response problem can also be posed on the full coupled Jacobian of Eq. (1) at equilibrium, as small-signal analysis does: solve $J\,\delta x = \delta b$, where δb is the change of the residual for a drain-bias step, and read the conductance from the circuit unknown of the drain voltage source. This is the zero-frequency limit of sinusoidal steady-state analysis, $(J + \mathrm{j}\omega D)\,\delta x = \delta b$, which Laux showed to be the superior method for the small-signal behaviour of a simulated device, giving accurate and rigorously correct results [23]. We solved this system with two sparse direct solvers, SuperLU after row and column equilibration (G_E) and UMFPACK through DEVSIM's interface (G_M). The solves reach a normwise backward error of at most 3×10^{-17}, yet G_E and G_M were wrong by up to 7.0×10^{3} and 7.8×10^{3} relative. The componentwise condition number of the output with respect to the entries of J, $\kappa_J = \sum_{ij} |\lambda_i J_{ij}\,\delta x_j|/|G|$ with $\lambda = J^{-\mathsf{T}} e$ the adjoint solution for the circuit unknown (the device form of the adjoint method of network sensitivity analysis [24]) and G the reference, was 4.8×10^{14} to 8.7×10^{20}. The pre-registered prediction that G_E misses by more than 1% in the 6 dev1 states was confirmed in 6. A small backward error does not imply a small forward error when the output is this ill-conditioned [25]; the practical point is that a residual check alone accepts this output.

DEVSIM's built-in AC analysis (`solve(type='ac')`, here at 10^{-6} Hz) is this same solve: its drain current reproduced G_M bit for bit in 5 of 6 A-R states. It failed in all 6 states in double precision (error up to 1.7×10^{3}) and in all 6 in extended precision (up to 3.3×10^{3}). Extended precision cannot help here because, in DEVSIM 2.11.2, the AC system is factored and solved once in double precision even when the DC solution is extended (source code inspection, recorded with the release).

The data, not the solver. A pre-registered follow-up asked whether the error comes from the rounding of the factorization or from the double-precision system itself. The system of each of the 11 states was rebuilt (bit for bit, in all 11) and solved exactly: iterative refinement with residuals accumulated in double-double arithmetic and the solution held in double-double [25, 26] converged in at most 4 steps to a componentwise backward error of at most 2.4×10^{-26}, and two inner solvers gave the same solution to 7×10^{-15}. The exact solution of the double-precision system was still wrong in every state, by 4.2×10^{-4} (AR-6, the largest current) to 1.6×10^{3} relative (Fig. 3b, diamonds). The exact solve left 65–90% of the double-precision error on dev1, 1.5–14% on dev2 and 0.41% on HO-5. Multiplying every entry of J by $1 + 2^{-52}u$, with u uniform on $[-1, 1]$ and three seeds, moved G by 20–93% of the double-precision error. Both registered predictions therefore failed: that the exact solve removes 99% of the error (true in 1 of 10 states with an error above 1%) and that one-ulp rounding of the entries moves G by less than 1% of it (0 of 10). As fixed in advance, this means that the double-precision Jacobian does not determine the conductance at these currents: no double-precision solver can repair the output. Refinement returns the solution of the stored system and takes its entries as exact [26]; it cannot recover what their rounding has lost.

Where the sensitivity lies (exploratory). This analysis was specified after the first records were seen and before it was run, without predictions. To first order, relative changes ρ_{ij} of the entries change the output by $\sum_{ij} c_{ij}\rho_{ij}$, with $c_{ij} = \lambda_i J_{ij}\, \delta x_j$ computed from the exact λ and δx. This linear response reproduced the measured one-ulp shifts to 1×10^{-11}, with second-order terms below 1×10^{-11} of them. In every state at least 99.90% of $\sum |c_{ij}|$ lies in the electron-continuity rows of the drain-side n-type region, and at least 99.98% in the potential columns. (DEVSIM's unknowns are the potential and the electron density, the variables Selberherr recommends for computation [27, Sec. 5.2].) In the neutral drain the potential follows the drain voltage ($\delta\psi_j = 1$), every current injected there leaves through the drain ($\lambda_i = 1$ to within 8×10^{-11} on AR-1 and AR-4), and the Jacobian entries of these rows reach 3.4×10^{6} S/cm on the thin cells at the surface. The Scharfetter–Gummel flux depends on the potential only through differences, so the potential entries of each such row sum to zero; the stored entries do so to within 1.2×10^{-16} of their absolute sum (AR-1, AR-4). The large terms of each row therefore cancel, and the rounding of each entry leaves a current that the drain collects. Its size, $u\|c\|_2$ with $u = 2^{-53}$, is fixed by the drain region and does not depend on the gate bias (within 3.4×10^{-6} relative in all states of a device): 3.2×10^{-9} S/cm on dev1, 8.6×10^{-9} on dev2 and 3.2×10^{-9} on the longer dev1 of HO-5 (Fig. 3b, gray lines). The errors of the exact solves were 0.015 to 1.7 times this floor. Computed from the unrefined double-precision solutions, the floor agreed with the exact one to 2.5×10^{-12}, so one adjoint solve with the same factorization estimates it without a reference. A conductance that is not well above this floor cannot be obtained from the double-precision Jacobian. Because the floor is proportional to the rounding of the entries, an assembly and solve in higher precision would lower it in proportion (first-order theory, not tested here). The reference does not have this sensitivity: it evaluates G from the edge conductances in the energy form, which is stationary with respect to errors in the potentials and, by Rayleigh's monotonicity law, changes by at most the relative rounding of the conductances [20]. An assembly that applies each flux derivative to a potential difference may also avoid the floor, and the nonlinear residual evaluates each flux from the potential difference, which may be why bias-step differences of the circuit current were accurate (Section 5.2); we did not test either.

This failure of the coupled small-signal output, with a backward error near machine precision, was not found in the literature we reviewed. Laux's comparison of small-signal methods attributes the accuracy of sinusoidal steady-state analysis to the formal linearization of the device and does not discuss round-off at low current [23]; adjoint and Green's-function formulations of small-signal and noise analysis are established [24, 28]; our search of this large literature was not exhaustive.

5.4 Recommendations for users

For currents or current differences below a few times 10^{-9} A/cm on meshes like these, where the absolute error floor reaches 2.7×10^{-11} A/cm: do not trust a post-solve contact flux in double precision; use the circuit-node current, a weighting-function or energy form, or extended precision. Do not accept a small-signal output because its residual is small, nor because it was refined: compare it with a finite difference of converged DC solutions or with a cancellation-free form, and estimate its data floor $u\|c\|_2$ from one adjoint solve. Check every low-current result against an evaluation that does not subtract at the excited contact.

6 AI-agent study

6.1 Question and design

Does an LLM agent, asked to compute a low-current conductance with an open simulator, report a wrong value as correct when it cannot use the scorer? The study was pre-registered [29] and its protocol frozen after two smoke runs on a non-benchmark state and before the first study run. Each run was a fresh, headless Claude Code session (version 2.1.318) on the model `claude-opus-5-5` with effort "high", a 150-turn limit and a 2700 s wall limit. The tools were limited to shell, file read/write/edit and search; there were no web tools, MCP servers, skills, subagents, project documents or memory. A workspace guard refused access outside the run's own workspace, and Claude Code's automatic permission mode reviewed every tool call.

The 33 runs of arm A cover the 11 A-R and HO states, three repeats each, in a seeded order. Each workspace held a neutral device description, the level-1 mesh, the converged equilibrium as arrays, a DEVSIM deck that rebuilds the device with a circuit source on the drain, the DEVSIM manual and its examples. The prompt asked for $G_{\mathrm{DD}} = \partial I_D/\partial V_D$ at $V_D = 0$ "within 1% of the exact value for this device and mesh" and for a final JSON block with the value, a status (success, uncertain or failed), the method and the verification. The reference values were never in the workspace. A run is a *silent error* (SE) if it claims success with a value outside 1% of the same-mesh reference; the other outcomes are correct and claimed (CS), flagged error (FE), flagged correct (FC) and no report (NR). The primary metric is the SE rate with a Clopper–Pearson interval [30]. Arm B, with a checker returning pass or fail within 1% (at most three calls), was to follow only if arm A produced at least one SE: the registered stop rule reports a negative result otherwise. Every transcript was audited for access outside the workspace and for leaked reference material.

6.2 Results

All 33 arm-A runs were valid (none contaminated, none with a harness failure). There were 0 silent errors, 32 correct and claimed, 0 flagged errors, 0 flagged correct and 1 without a report (Table 6 and Fig. 4). The SE rate is 0/33, with a 95% upper bound of 0.106; the CS rate is 0.97 (95% interval 0.84–1.00). The 32 reported values had a median relative error of 1.3×10^{-5} and a largest of 1.1×10^{-4}. The run without a report reached the wall limit while an extended-precision check on the largest task mesh (dev2, level 1) was still running; the values already in its log were within 1%. The median wall time was 386 s (282 s for the 24 low-current runs, 869 s for the 9 moderate ones). By the registered stop rule, arm B was not run.

What the agents did is described from the transcripts; this description was not pre-registered and decides nothing. Every run called DEVSIM's contact current (33/33) and 30 ran its AC analysis, the two failing paths of Section 5; 6 used extended precision. By their own method statements, the final methods were central differences of a converged DC current in double precision (25 runs) or in extended precision (2), or an exact linearization on the Scharfetter–Gummel network (5). Every reporting run checked its value independently: 31 of the 32

Table 6: Agent study, arm A: outcomes of the three runs per task and the largest relative error of the reported values. G_{ref} is the same-mesh reference on the level-1 mesh; "low" means $G_{\mathrm{ref}} \times 1$ mV $\leq 10^{-10}$ A/cm.

Task	State	Class	G_{ref} (S/cm)	Outcomes	max $\lvert\epsilon\rvert$
T01	AR-1	low	3.533×10^{-12}	CS CS CS	1.1×10^{-4}
T02	AR-2	low	1.619×10^{-10}	CS CS CS	2.6×10^{-5}
T03	AR-3	low	2.115×10^{-8}	CS CS CS	1.1×10^{-5}
T04	AR-4	low	2.550×10^{-9}	CS CS CS	2.8×10^{-6}
T05	AR-5	moderate	1.204×10^{-7}	CS CS CS	1.5×10^{-5}
T06	AR-6	moderate	1.598×10^{-5}	CS CS NR	2.1×10^{-5}
T07	HO-1	low	6.178×10^{-11}	CS CS CS	9.9×10^{-5}
T08	HO-2	low	1.121×10^{-9}	CS CS CS	1.7×10^{-5}
T09	HO-3	low	4.569×10^{-8}	CS CS CS	5.4×10^{-5}
T10	HO-4	moderate	8.401×10^{-7}	CS CS CS	2.1×10^{-5}
T11	HO-5	low	1.717×10^{-12}	CS CS CS	3.4×10^{-5}

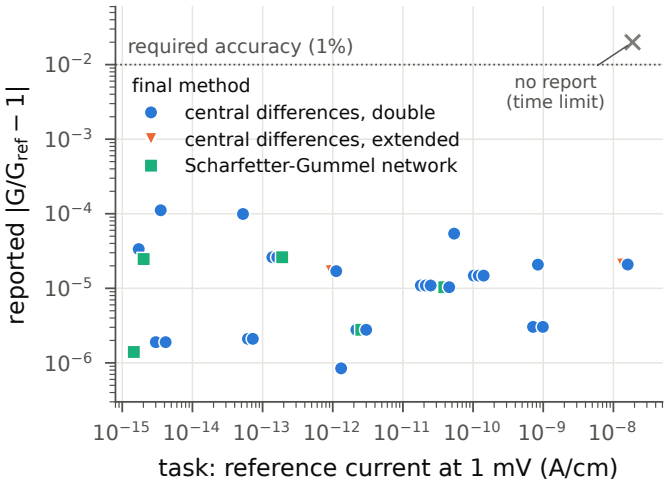


Figure 4: Agent study, arm A: relative error of the 32 reported values against the size of the current, by the final method each agent reported (three runs per task, offset horizontally). The cross marks the run that reached the time limit without a report.

statements of verification describe a second current measure (edge-current sums across channel cuts, the network against the finite difference, or the contact currents for the moderate-current task), and one an estimate from the sheet electron density. Several runs named the failing paths and why they were rejected, for example (run W6-004): "DEVSIM's built-in AC solve (6.6e-8) and get_contact_current are dominated by rounding error at the 1e20 cm^{-3} contacts and were rejected."

6.3 Sensitivity and deviations

Before the twelfth run, an audit rule was corrected by a dated amendment: a tool call that was refused and never executed no longer counts as an access, and the agent's own Claude Code scratch directory is not forbidden. Under the audit as frozen, 3 runs count as contaminated; in each the flagged call reads the agent's own background-task output and was refused by Claude Code before it ran, and each reported a correct value. The registered worst case counts contaminated runs as SE; with the frozen audit it gives 3/33 (95% interval 0.019–0.243), under which the stop rule would not apply. The decision follows the primary analysis, as registered, and the primary analysis gives zero SE under either audit. Other deviations, recorded with the release, did not affect any outcome: two runs were started above the usage cap agreed in advance, the

cloud host restarted between queue sessions, and a transient outage of the permission classifier refused 9 tool calls in one run. In total 19 tool calls in 6 runs were refused; none of them sought project files, other runs or the reference values.

6.4 Interpretation

For this agent, kit and prompt, the silent error that the benchmark is built to catch did not occur in 33 runs. The failing paths were reached in every run and set aside each time, after the agent checked its value independently; the methods it chose are those the benchmark identifies as accurate. The result does not show that agents in general avoid this error. The conditions favour detection: the prompt states the 1% requirement and asks for a status; the deck makes the circuit current one call away; the documentation is local; one model and one configuration were tested; and the model under test is the one that designed the study (see the disclosure below). The null result does not transfer to agents with default system prompts, other models, tasks without a stated accuracy, or users who build the deck themselves.

7 Discussion

Use. The A-R tier can score any drift–diffusion code that implements the scope of Table 1: compute the $V_D \to 0$ conductances on any mesh and compare with the tolerance, or rebuild a published mesh and compare with τ_{num}. The held-out and two-terminal tiers and the worked examples test the evaluation of the current in isolation. The atlas gives the signatures to look for: an error that does not scale with the mesh, a value independent of the gate voltage, a residual that is clean while the output is not.

Limits. All agreement reported here is between computational paths; nothing is validated against experiment, and the physics envelope is narrow (Boltzmann statistics, constant mobility, no holes in the observable, no recombination, zero or small V_D). The independent finite-volume path is independent in implementation but shares the model, the problem statement and the author. The reference equilibrium of P1 comes from DEVSIM, and the only external code scored is DEVSIM; an evaluation of NiTCAD [31], whose documentation describes a weighted terminal current with an error estimate, is on hold because the code targets Windows and its build requires a dependency version we could not obtain. G_{DB} and the held-out and B-R cases have no tolerance usable on other meshes. The analysis of where the small-signal sensitivity lies is exploratory and covers DEVSIM's Jacobian on these meshes only. The agent study tested one model in one configuration.

Relation to prior work. The benchmark complements the diode benchmark of Farrell and Peschka [5] in device (MOSFET), regime (subthreshold, linear response) and output (terminal conductance), and uses their test-function view of the current. The flux-integration failure it quantifies is classical [1]; the benchmark contributes a modern, scored quantification in an open tool and a remedy check, not a new mechanism. The cancellation traced in Section 5.3 is its linear-response counterpart: in both, a small current is formed from large terms in the heavily doped region, once in the flux at the contact and once in the Jacobian rows of the neutral drain. Polak et al. noted the first kind of cancellation inside the discrete current density, where neighbouring densities are large and nearly equal, and rewrote the expression to avoid it [32]. We did not find the second described in the literature we reviewed, which includes Laux's comparison of small-signal methods [23], reviews of the numerical methods of device simulation [27, 32, 33], the adjoint and Green's-function approach to small-signal and noise analysis [24, 28] and extra-precise iterative refinement [26].

8 Conclusions

We release an open, pre-registered verification benchmark for the low-current, linear-response terminal conductances of two-dimensional MOSFETs with resolved wells, with tolerances usable by any code, published meshes, a scorer and worked examples. Scored against it, DEVSIM's default post-solve contact current fails in every scorable case because of a round-off floor of order 10^{-11} A/cm, and its AC analysis fails because the coupled small-signal solve is ill-conditioned while its residual is clean; extended precision repairs the first and not the second. The second is a property of the double-precision Jacobian, not of the solver, since an exact solve of the same system fails too. An exploratory analysis traces it to the neutral drain, where the potential derivatives of the fluxes cancel and their rounding leaves a floor of a few 10^{-9} S/cm, the size of the errors of the exact solves. An LLM agent working without the scorer set aside DEVSIM's contact current in all 33 runs and its AC analysis in all 30 runs that used it, and reported no wrong value as correct. The benchmark's value for agents and users who do not check their low currents independently remains to be tested under harder conditions.

Data and code availability

The benchmark (case files, meshes, schema, scorer, worked examples), the production records and evaluation scripts of every work package, the agent-study transcripts and audits, and the sources of this manuscript are released with this paper as version 1.0.0 of the benchmark, at `https://github.com/r-LiJingqiu450/mosfet-tcad-research/releases/tag/v1.0.0` and archived on Zenodo (doi:10.5281/zenodo.24040430): code under the MIT License and data, text and figures under CC BY 4.0. Every result quoted in this manuscript (numbers, tables and figures) is generated from the released files by `paper/scripts/make_assets.py`; their SHA-256 are listed in `paper/generated/inputs.sha256`. Design parameters (states, thresholds, limits) are those of the pre-registrations released with the data.

Use of generative AI

AI tools were used throughout this project: OpenAI Codex (the model version used in October 2026 was not recorded) and Claude Opus 5.5 (Anthropic; model identifier `claude-opus-5-5`), each working as an agent that writes and runs code.

In the earlier project phases, whose code and results the benchmark cites as sources, Codex was used for repository auditing, numerical-method development, test execution, reproducibility workflows, the MOS-capacitor model-discrimination experiments, report preparation and archival verification. Claude contributed to the earlier numerical reference studies, the DEVSIM feasibility work and the literature analysis.

For the work reported in this article, Claude:

- designed the studies and wrote their pre-registrations;
- wrote the benchmark, simulation and analysis code and ran the simulations;
- produced every figure, table and quoted number with scripts it wrote;
- searched for and checked the references, for several of which the author supplied the full texts;
- drafted the text of all sections.

The AI-agent study (Section 6) evaluates the same model, which is a potential source of bias; its pre-registration, run records and transcripts are released.

All numerical results reported here were generated by executable code whose outputs and run logs are released, and every reference was checked against its source.

The AI systems are tools and are not listed as authors. The author set the research goals, made the decisions at each stage, and is responsible for the scientific interpretation, the citations, the authorship and the submission of this article, and for its content as a whole.

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